

A large-scale systematic map of anthropogenic drivers of biodiversity loss

Hunain Mohuiddin¹, Kerstin Forster^{1,2}, and Stefan Feuerriegel^{*,1,2}

¹LMU Munich, Munich, Germany

²Munich Center for Machine Learning, Munich, Germany

*Corresponding author: feuerriegel@lmu.de

Abstract

Biological diversity underpins ecosystem functioning and planetary resilience. Biodiversity research has expanded rapidly, yet keeping track of which anthropogenic drivers, ecosystems, and taxa are most studied remains a major challenge. Here we present a large language model (LLM)-based framework to systematically map the evidence base on anthropogenic drivers of biodiversity loss at scale. Our framework classifies scientific publications by the anthropogenic drivers, specific threats, geographic scope, ecosystems, study designs, and taxa involved, drawing on established biodiversity classification schemes; taxonomic identity is resolved against a comprehensive reference covering over two million described species. Our LLM-based mapping yields a systematic map of 253,081 scientific articles on biodiversity loss, revealing disparities in research attention across drivers, ecosystems, and geographies. This map offers a foundation for identifying research gaps, tracking progress, and guiding priorities in biodiversity science and conservation.

15 **Main**

16 Achieving the 2050 Vision of living in harmony with nature, set out in the Kunming–
17 Montreal Global Biodiversity Framework (KM-GBF), requires transformative changes
18 across economic, social, political, and technological systems [1]. Central to this trans-
19 formation is understanding what drives biodiversity loss.

20 Biodiversity research is highly heterogeneous: it spans an enormous range of taxa,
21 ecosystems, and geographic regions, and employs diverse methodological approaches
22 across exposures as varied as land-use change, pollution, invasive species, climate change,
23 and direct exploitation. Crucially, these drivers do not act in isolation: land-use pres-
24 sures interact with climate change, overexploitation compounds habitat degradation, and
25 conservation outcomes depend on which combination of drivers is in play [1, 2]. Any
26 synthesis of the global evidence base must grapple with this breadth and interdependency
27 simultaneously.

28 Yet the existing literature remains structured by pronounced imbalances. First, a small
29 subset of taxa receives the vast majority of research attention while most species remain
30 understudied [3–6]. Second, driver coverage is similarly skewed: climate change accounts
31 for nearly 45% of driver-related publications, while pollution appears in fewer than 5% [7].
32 Third, existing systematic maps and meta-analyses address a single driver, taxon, or re-
33 gion, reflecting these same biases [8–11]. A comprehensive synthesis that jointly charac-
34 terises the evidence base across all major drivers, specific threats, ecosystems, geographic
35 context, and taxa is therefore absent.

36 Here, we present such a map, a structured overview of over 253,081 scientific articles

on the direct anthropogenic drivers of biodiversity loss, retrieved from the Web of Science Core Collection (WOS) and classified across six dimensions: anthropogenic drivers, specific threats, geographic scope, ecosystems, study designs, and taxa.

Classifying evidence at this scale is infeasible manually. We therefore automate the screening and coding pipeline using Large Language Models (LLMs) following Collaboration for Environmental Evidence (CEE) systematic map guidelines [12, 13]. The result is a structured, comprehensive evidence base that captures the full breadth of biodiversity research and can be continuously updated as new literature accumulates.

In this article, (preview results summary: TBA at last)

Results

Scientific evidence on biodiversity loss is extensive and rapidly expanding

Our systematic map comprises 253,081 scientific articles reporting biodiversity loss linked to direct anthropogenic drivers. Each article is coded across six dimensions: the five IPBES direct drivers, specific IUCN threats, geographic scope, ecosystem context, study design, and taxonomy. Together, these classifications form a structured, multidimensional evidence base in which research attention can be analyzed jointly across drivers, places, ecosystems, methods, and taxa.

The map reveals that the annual number of scientific articles on biodiversity loss increased from 2,530 in 2000 to 20,520 in 2025, a compound annual growth rate (CAGR) of 8.73% (see Figure 1a). Growth was especially pronounced for climate change and severe weather, which increased at 14.75% annually compared with the overall rate; the cate-

58 gory's share of broad threat attributions rose from 5.3% to 19.3% (see Figure 1b,c). This
59 increasing research emphasis is consistent with previous evidence that scientific attention
60 is unevenly distributed across biodiversity-loss drivers and particularly concentrated on
61 climate change [7]. However, these proportions describe the distribution of scientific at-
62 tention, not the relative contribution of each driver to biodiversity loss. The systematic map
63 provides the common evidence base for the analyses that follow, which examine how re-
64 search attention is distributed across drivers, geographies, ecosystems, and socioeconomic
65 contexts.

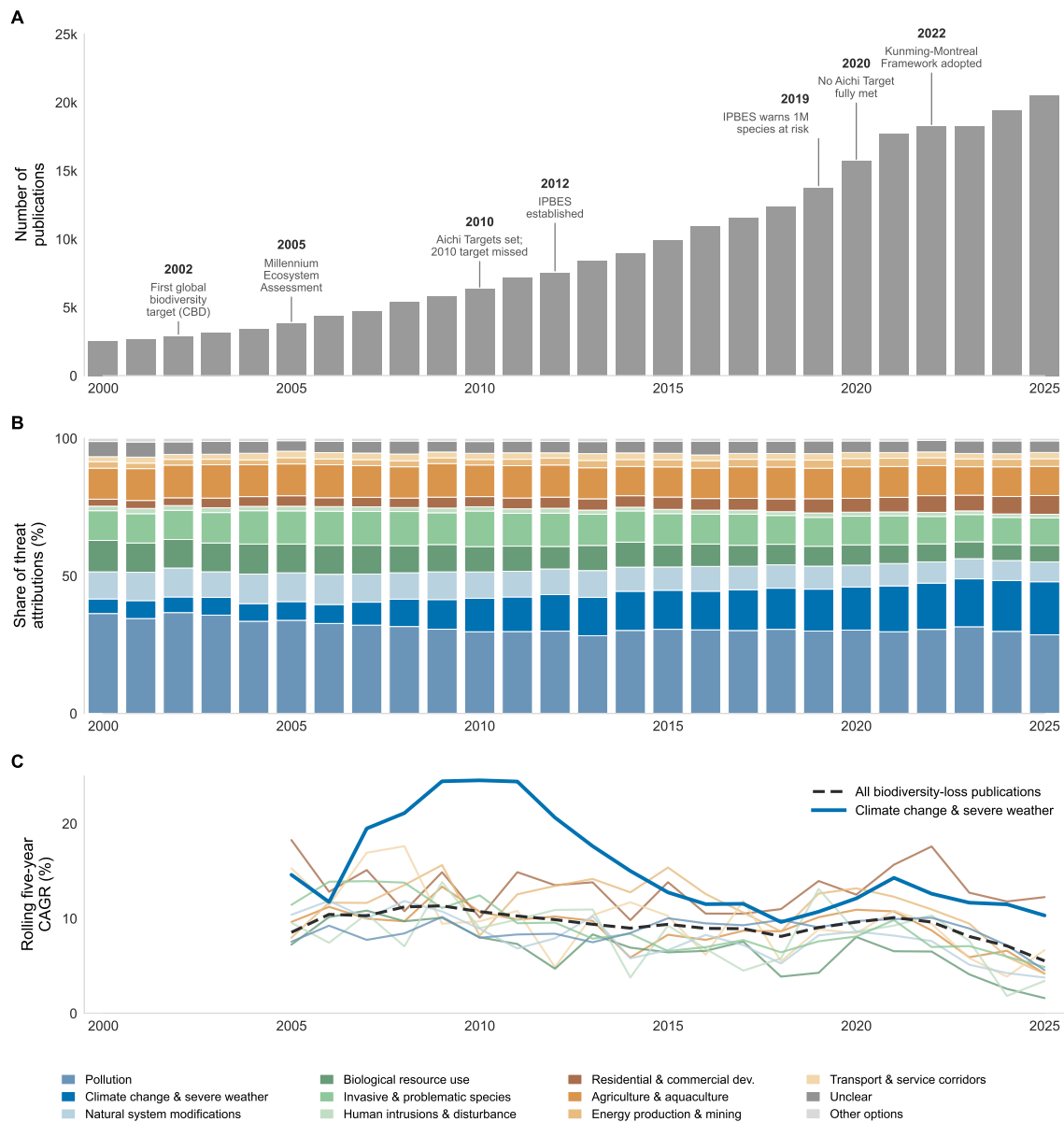


Figure 1: Temporal development and changing composition of the scientific evidence on biodiversity loss: (a) Annual number of scientific articles for complete publication years from 2000 to 2025. Selected international biodiversity-policy and assessment milestones are shown for context. (b) Annual distribution of broad IUCN threat categories, expressed as the percentage of all threat attributions assigned in each year. (c) Five-year CAGR for the complete biodiversity-loss evidence base and individual broad threat categories.

Biodiversity-loss evidence differs systematically across national income groups

Global assessments identify land- and sea-use change and direct exploitation as the two direct drivers with the largest impacts on nature to date [2]. The scientific literature, however, documents these drivers unevenly across national income groups. Across 94,443 scientific articles reporting observational evidence of biodiversity loss in a specific country, the documented-threat composition of lower-income-country evidence was dominated by agriculture and aquaculture and by biological resource use, whereas higher-income-country evidence gave greater relative emphasis to biological invasions and to climate change and severe weather. The largest contrasts were for agriculture and aquaculture (16.0 percentage points lower in high-income evidence; 95% bootstrap confidence interval 15.0–17.0) and biological resource use (12.2 points lower; 11.2–13.3), alongside biological invasions (9.1 points higher; 8.3–9.9) and climate change (6.0 points higher; 5.2–6.7) (see Figure 2).

These contrasts persisted after directly standardizing the evidence to a common regional and publication-period composition, indicating that they were not explained by differences in where and when the scientific articles were produced: after joint standardization, lower-income-tier evidence remained more oriented toward agriculture (7.0 points) and biological resource use (5.4 points), and higher-income-tier evidence toward biological invasions (5.2 points) and climate change (3.0 points). The pattern held in direction and rank across alternative study-design, income-classification, and weighting specifications (see Supplementary Results). The prominence of agriculture and direct resource use in lower-income-country evidence is consistent with global spatial analyses showing that agriculture and logging are particularly pervasive across tropical regions, while

89 hunting and trapping are geographically widespread threats to mammals and birds [14],
90 and may also reflect the organization of global production, in which consumption in
91 wealthier economies is linked to biodiversity threats in developing producer countries [15].
92 The greater relative representation of biological invasions and climate change in higher-
93 income-country evidence may, in turn, partly reflect persistent geographic inequalities in
94 biodiversity research, with studies disproportionately concentrated in developed countries
95 with greater economic and research capacity [16]. These profiles describe documented
96 research attention, not the ecological prevalence or severity of threats.

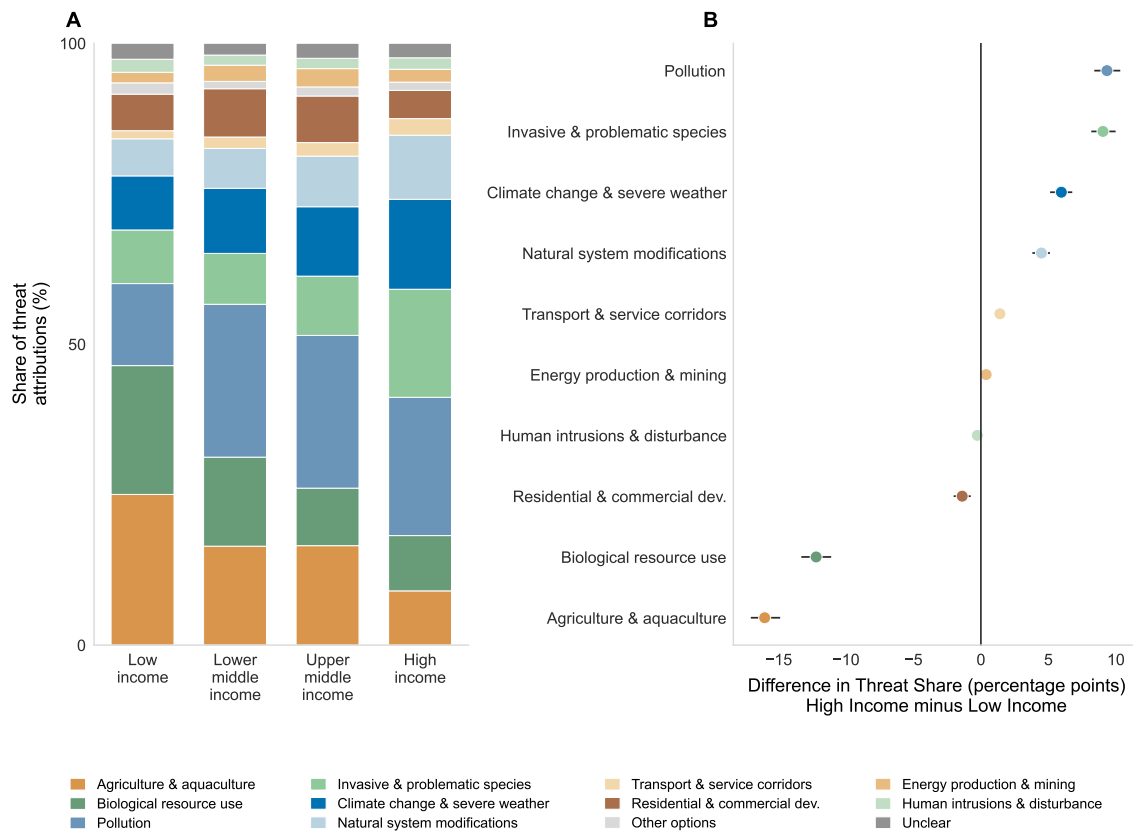


Figure 2: Documented-threat composition of the biodiversity-loss evidence differs across national income groups: Threat attributions are drawn from 94,443 unique publications reporting negative-direction, observational evidence linked to a country of study, with each publication's attributions allocated fractionally across the broad IUCN threat categories it reports. **(a)** Within-group composition, expressed as each threat category's share of all attributions in a given World Bank income group; income group is matched to the publication year. **(b)** High-income-minus-low-income difference in each category's attribution share, in percentage points; points show the observed difference and horizontal lines the 95% bootstrap confidence interval, ordered by the absolute size of the contrast. Differences shown here are unadjusted; contrasts standardized to a common regional and publication-period composition are reported in the Supplementary Results.

97 **The misalignment between research attention and driver impact is consistent across**
98 **ecosystem realms**

99 Global assessments hold that the direct drivers of biodiversity loss differ in importance
100 across the terrestrial, freshwater, and marine realms, with land-use change foremost on
101 land and in freshwater and direct exploitation foremost in marine systems [2, 10]. We ex-
102 amined how the five IPBES direct drivers were distributed across scientific articles within
103 each realm (Figure 3). In every realm, documented research attention departed from this
104 hierarchy in the same direction: pollution drew more attention, and direct exploitation less,
105 than their assessed impact would imply. Pollution held the largest share in both freshwa-
106 ter (57%) and marine (38%) evidence, while direct exploitation stayed minor everywhere,
107 despite being the most prevalent documented threat to species worldwide, affecting 72%
108 of threatened and near-threatened species [17], and the leading assessed marine driver [2].
109 Only on land did the assessed leading driver, land-use change (33%), also lead in attention.

110 The freshwater realm shows this most clearly. Its biodiversity loss is attributed to
111 five interacting pressures of comparable standing, including flow alteration and habitat
112 degradation alongside pollution [18], yet freshwater attention concentrated on pollution
113 more heavily than in any other realm. Where the real causes are many and diffuse, the
114 evidence narrows onto the one easiest to identify and measure. The gap is also widening:
115 attention to direct exploitation fell in every realm over the study period, most steeply in
116 marine evidence, even as it remained the leading assessed marine driver (Supplementary
117 Figure S4).

118 These patterns show that documented research attention does not follow assessed im-
119 pact, departing from it in the same direction across all three realms. This extends a bias

documented across taxa and regions [3, 16] to the direct drivers of loss themselves, revealing a systematic gap between the drivers most represented in the evidence and those assessed to matter most [19].

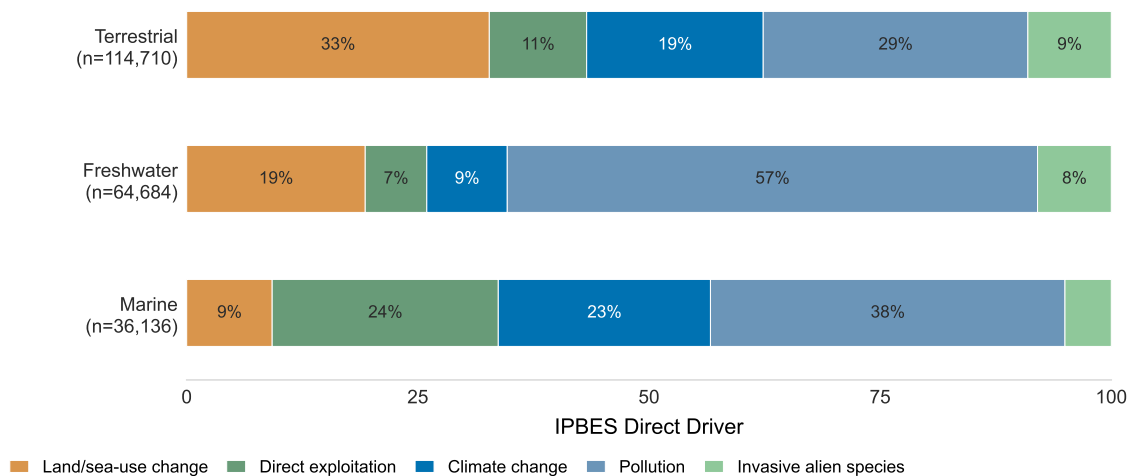


Figure 3: The composition of the biodiversity-loss evidence differs across ecosystem realms: Each bar shows the share of driver attributions accounted for by the five IPBES direct drivers within a realm, based on the scientific articles assigned to that realm (*n* shown per realm); each article’s attributions are allocated fractionally across the drivers it reports. Pollution holds the largest share in both freshwater (57%) and marine (38%) evidence, whereas direct exploitation remains a minor share in every realm despite being the most prevalent assessed threat to species and the leading assessed marine driver. Land-use change leads in attention only in the terrestrial realm (33%).

The drivers assessed to matter most are studied through the narrowest taxonomic lens

Research attention is known to favor vertebrates over invertebrates [3, 16].

Our map reconfirms this skew at scale: across 171,791 biodiversity-loss articles with a resolvable taxonomic group, vertebrates received 34.8% of documented attention against

128 4.6% of accepted described species (7.5 times), whereas invertebrates received 19.7%
129 against 65.0% (0.30 times). This skew, however, is not a fixed constant: it depends on
130 which driver of loss is under study. The vertebrate share of attention ranged from 25.9%
131 under climate change to 50.5% under direct exploitation, a difference of 24.5 percent-
132 age points (95% bootstrap confidence interval 23.9–25.2). Direct exploitation was the
133 only driver that over-represented vertebrates while under-representing all seven other tax-
134 onomic groups; pollution was its mirror image, the only driver to over-represent all four
135 invertebrate, fungal, and microbial groups.

136 The narrowest lenses sit on the drivers assessed to matter most. Land- and sea-use
137 change and direct exploitation are the two drivers assessed to have the largest impacts on
138 nature to date [2]. These are also the two whose evidence is most concentrated on verte-
139 brates. Pollution, by contrast, is the most taxonomically inclusive: it draws disproportion-
140 ate attention relative to its assessed impact across ecosystem realms, as shown above. The
141 gap therefore compounds rather than compensates: where the evidence base is thinnest
142 relative to assessed importance, it is also narrowest taxonomically. Growth has not cor-
143 rected this. Annual publication volume grew 8.11-fold between 2000 and 2025, yet the
144 vertebrate share of animal-taxon attention moved by only -0.24 percentage points, from
145 62.35% in 2000–04 to 62.11% in 2020–25 (95% CI -1.51 to $+0.98$): publication growth
146 has enlarged the evidence base without widening its taxonomic lens.

147 These patterns extend a taxonomic bias documented in prior work [3, 16] to the driver-
148 specific structure of the evidence base itself. The map describes documented research
149 attention, not ecological importance, conservation need, or the number of species affected
150 by a driver. It also cannot, on its own, distinguish an attention bias from real differences

¹⁵¹ in the taxonomic composition of what each driver affects, a distinction we return to in the
¹⁵² Discussion.

153 Discussion

154 The four analyses trace one evidence base along three independent axes of research atten-
155 tion: national income group, ecosystem realm, and taxon. Along each axis, documented
156 attention departs from assessed importance, and does so in a consistent direction: pollu-
157 tion research is disproportionately large relative to its assessed impact, while research on
158 land- and sea-use change and direct exploitation, the two drivers assessed to matter most,
159 is comparatively narrow. Because the map codes these axes jointly rather than in isola-
160 tion, it also shows that the resulting gaps compound rather than cancel: the drivers whose
161 evidence is thinnest relative to assessed importance are the same drivers whose evidence
162 is most concentrated on a single realm and a single taxon. A synthesis built on any one
163 dimension alone would have missed this stacking; a synthesis built on all six together is
164 what makes it visible.

165 This closes a gap our Introduction identified directly: existing systematic maps and
166 meta-analyses on biodiversity loss each address a single driver, taxon, or region [8–11],
167 and none jointly characterizes the evidence base across drivers, threats, geography, ecosys-
168 tems, and taxa at this scale. At 253,081 classified articles, our map is, to our knowledge,
169 the largest evidence base of its kind for biodiversity loss, and its value lies less in its size
170 alone than in what joint coding of six dimensions makes visible: compounding gaps that
171 single-dimension reviews cannot detect, because no single dimension shows them. A map
172 of this kind offers a layer of evidence beneath future synthesis and assessment work, one
173 against which the coverage of individual reviews, or the citation practices of assessment
174 bodies, could in principle be checked.

175 Three limitations qualify how these results should be read. First, the classification
176 pipeline, while validated, is not error-free, and its reliability varies by task: search com-
177 prehensiveness reached 85.54% recall against an a priori test list, screening recall reached
178 90.45% on the held-out test split, and coding performance averaged 83.54% macro-F1
179 (range 57.52–98.90%) and Cohen’s κ of 0.72 (range 0.25–0.93) across the six label sets
180 (see Supplementary Methods for the full validation results). Findings resting on less-
181 validated labels, such as ecosystem sub-typology or species-level taxonomy, carry propor-
182 tionally more uncertainty than those resting on driver, geographic, or genus-level labels,
183 and should be weighted accordingly (Supplementary Table S10). Second, the analyses pre-
184 sented here narrow progressively from the full evidence base: screening identified 605,199
185 eligible records reporting biodiversity change of any direction, of which 253,081 reported
186 loss and form the analyzed dataset, and individual findings draw on further subsets, such
187 as the 94,443 articles with a documented country of study underlying the income analysis.
188 Third, all compositional and temporal analyses are bounded to complete publication years,
189 2000 through 2025; typical indexing lag means even the final year is likely a slight under-
190 count, and drivers or taxa with recently accelerating attention are the most susceptible to
191 this.

192 A further caveat applies specifically to the attention-versus-importance comparisons.
193 Departures from assessed impact are consistent with under-attention, but the map cannot
194 fully separate that explanation from genuine differences in what each driver, realm, or
195 income context actually affects: exploitation-related research may concentrate on verte-
196 brates partly because fisheries and wildlife harvest genuinely target vertebrates more than
197 other groups, and pollution research may draw on invertebrate and microbial model organ-

isms for established ecotoxicological and regulatory reasons rather than because attention is unusually inclusive. This ambiguity is not unique to this map; it is inherent to any evidence-attention analysis that lacks an independent benchmark for what a driver actually affects, and it also applies to prior work documenting taxonomic bias [3, 16]. Only the taxonomic anchor offers such a benchmark here, against described species richness, and even that benchmark is suggestive rather than conclusive, since described diversity is not the same as what a specific driver targets. Testing the alternative directly would require comparable external benchmarks for the other axes, such as catch and harvest composition for direct exploitation or standard test-organism conventions for ecotoxicology.

These limitations narrow how strongly the findings should be read, not what they are useful for. The map points to where new evidence would close the largest documented gaps relative to assessed importance: non-vertebrate impacts of direct exploitation and land- and sea-use change, and driver evidence beyond pollution in freshwater and marine systems, whose actual pressures are documented as more numerous and interacting than the literature reflects [18]. It also distinguishes evidence that is already rich but has not been synthesized from evidence that remains genuinely thin, a distinction future systematic reviews could use to prioritize where synthesis, rather than further primary research, is the more efficient next step. More broadly, a jointly coded map of this kind offers a check available to assessment bodies such as IPBES: whether the evidence cited in an assessment tracks the underlying distribution of research attention, or draws disproportionately on its most convenient or most-cited subset. Resolving the attention-versus-importance ambiguity raised above, through external benchmarks for exploitation and pollution targets analogous to the taxonomic anchor used here, is the most direct next step this map

221 motivates.

222 **Methods**

223 **Systematic mapping protocol**

224 We used an LLM to create a comprehensive systematic map of direct anthropogenic drivers
225 of biodiversity loss, following CEE guidelines for systematic mapping [12] with adjust-
226 ments for our LLM-based pipeline. All steps are documented in a protocol pre-registered
227 prior to full-corpus application. Figure X summarizes the five-stage workflow, compris-
228 ing record retrieval, label acquisition, prompt definition for screening, prompt definition
229 for coding, and full-corpus application, each detailed in the following subsections. LLM-
230 specific configurations and rationales for each stage are documented in an accompanying
231 reporting checklist [20] (Extended Data Table 1).

232 **Document search**

233 The first stage retrieved bibliographic records from the WOS. We constructed a search
234 string designed to prioritize recall, combining driver-specific terms adopted from a prior
235 evidence synthesis [10] with a broad set of biodiversity-relevant subject categories [9]
236 (see Supplementary Table S2). Comprehensiveness was assessed against an *a priori* test
237 list of 581 records drawn from five published evidence syntheses [9, 10, 21–23] and the
238 IPBES Global Assessment chapters on direct drivers [24], yielding an overall recall of
239 85.54% (see Supplementary Table S3). Applied to publications from 2000 to April 2026,
240 the search retrieved approximately 2.4 million bibliographic records after deduplication.

Document screening through LLM

The second stage screened the retrieved records for evidence of a biodiversity change linked to a direct anthropogenic driver. We implemented screening as a zero-shot classification task using GPT-5-nano (version 2025-08-07): each record’s title and abstract was assessed through a four-question procedure (see Figure S1), and a record was classified as eligible only when all four questions returned `Yes` or `Unclear`. To assess reliability, we compared LLM screening decisions against expert judgments from previously published, curated evidence syntheses [10,25–28]. A human baseline, established by having a person screen from abstracts alone ($n = 100$), achieved a recall of 83%. The LLM, evaluated against the full-text expert labels ($n = 2,418$, partitioned into training, development, and held-out test splits), achieved a recall of 90.45% on the test split ($n = 482$). A further check compared LLM and human screening decisions on 400 records drawn directly from search retrieval rather than this curated benchmark, finding a recall of 90.1% (precision 64.5%, F1 75.2%), consistent with the LLM’s performance on the curated data. Applied to the full corpus, screening yielded 605,199 eligible records, of which 253,081 reported a negative direction of change and constitute the biodiversity loss dataset analyzed in this study (Supplementary Figure S2).

Document coding through LLM

The third stage coded eligible records across six label sets (L1–L6) that together reconstruct an evidence chain from direct anthropogenic driver to biodiversity impact: driver category (L1), specific threats (L2), geographic scope (L3), ecosystem typology (L4), study attributes (L5), and taxonomy (L6) [19,29–32]. Label definitions and decision rules

were adopted verbatim from these established schemes; L5 follows a custom schema developed for this map. Hierarchical label sets (L2, L4) required multiple LLM calls per record to resolve their threat or ecosystem hierarchy; all others used a single call. As with screening, we assessed reliability by comparing LLM coding decisions against expert judgments: a human baseline established how well a person could code from abstracts alone, and a second check evaluated the LLM against full-text expert labels across all six label sets. On the held-out test split, the pipeline achieved a macro-average weighted F1 of 83.54% (range: 57.52%–98.90%) and Cohen’s κ of 0.72 (range: 0.25–0.93).

Validation

Both screening and coding used GPT-5-nano (version 2025-08-07) throughout, selected after comparison against larger alternatives where marginal performance gains did not justify cost at the scale of this map. Reasoning effort was the only varied parameter, selected per label set on the development split. Both reference datasets were partitioned into training, development, and test splits; prompts were developed on the training split, reasoning effort selected on the development split, and the test split held out for final evaluation (see Supplementary Methods).

279 **Data availability**

280 Data for all 253,081 articles on biodiversity loss coded in this systematic map are avail-
281 able via Open Science Framework (OSF) at <https://osf.io/xg8yq/overview>. WoS-derived
282 metadata is excluded in compliance with Clarivate’s end-user terms; full details on record
283 identifiers, licensing restrictions, and reproduction instructions are documented within the
284 OSF repository. **Review once before submission: Add final .csv data in repo along with**
285 **instructions**

286 **Code availability**

287 All code used for the LLM-based screening and coding pipeline and validation is available
288 at <https://github.com/hunaindot/biodiversity-taxonomy-thesis/tree/main>. The repository
289 includes prompts, configuration files, and a README detailing setup and reproduction
290 steps.

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384 reports (2017). Version 1.0.

385 **Acknowledgments**

386 We thank the authors of the published evidence syntheses used to construct the search test
387 list and the validation datasets for screening and coding of this systematic map

388 **Author contributions**

389 H.M. designed the study, developed the LLM-based screening and coding pipeline, per-
390 formed the analysis, and wrote the manuscript. K.F. reviewed the manuscript and pro-
391 vided feedback. S.F. conceived the study, provided supervisory guidance, and reviewed
392 the manuscript. All authors read and approved the final manuscript.

393 **Funding**

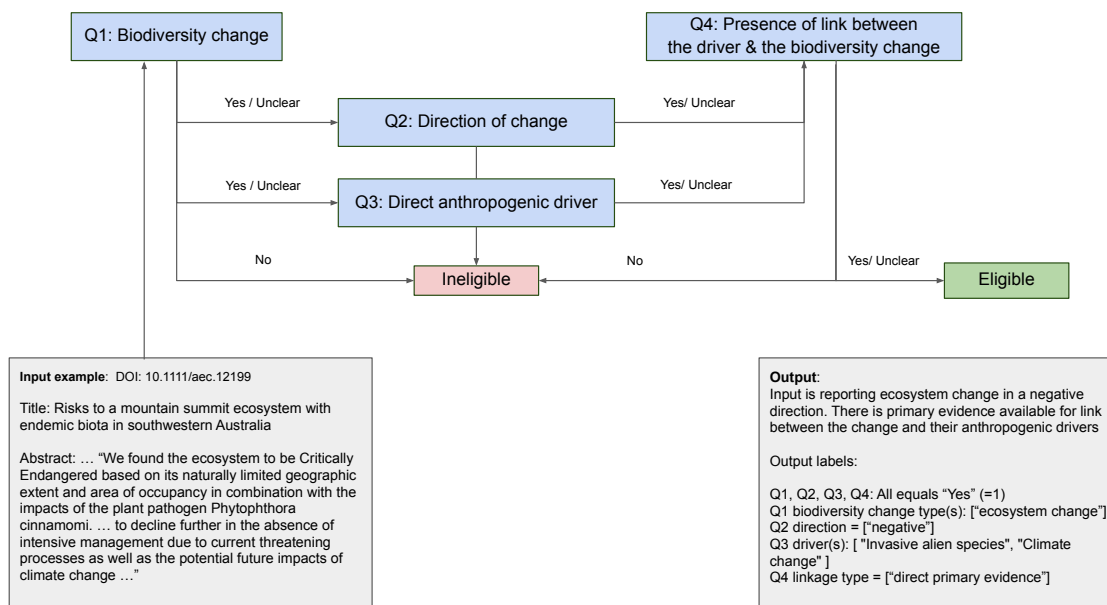
394 No external funding was received.

395 **Competing interests**

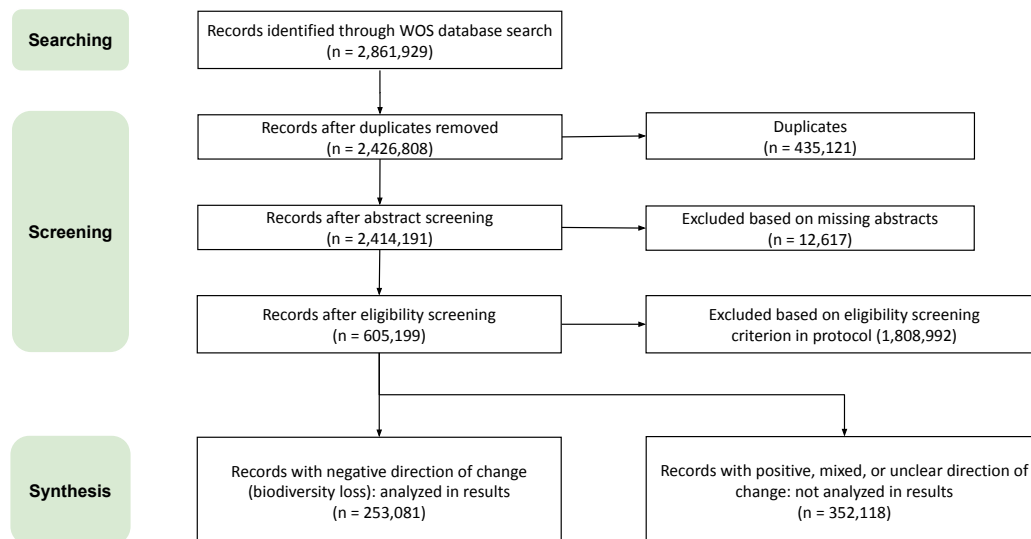
396 The authors declare no competing interests.

397 **Supplementary information**

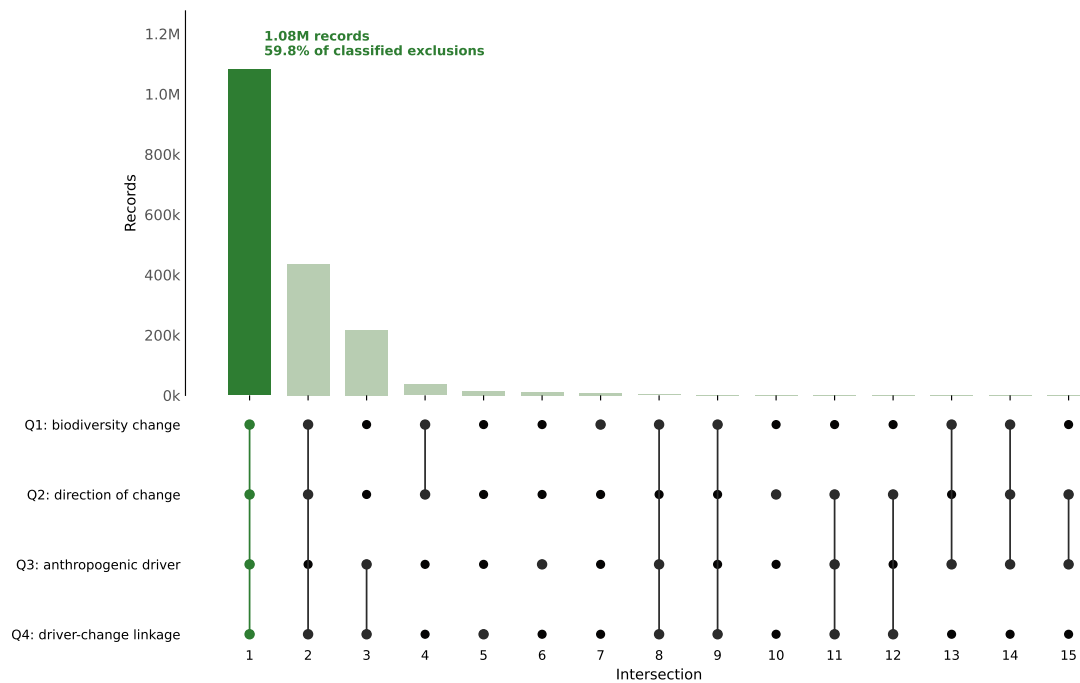
398 This section provides supporting material for the results and methods summarized in the
399 main text: Supplementary Tables, Supplementary Figures, and Supplementary Methods
400 detailing the pipeline underlying the main-text results. All materials required to reproduce
401 these methods are available via the registered protocol, code repository, and data repository
402 **Review once again before final submission**



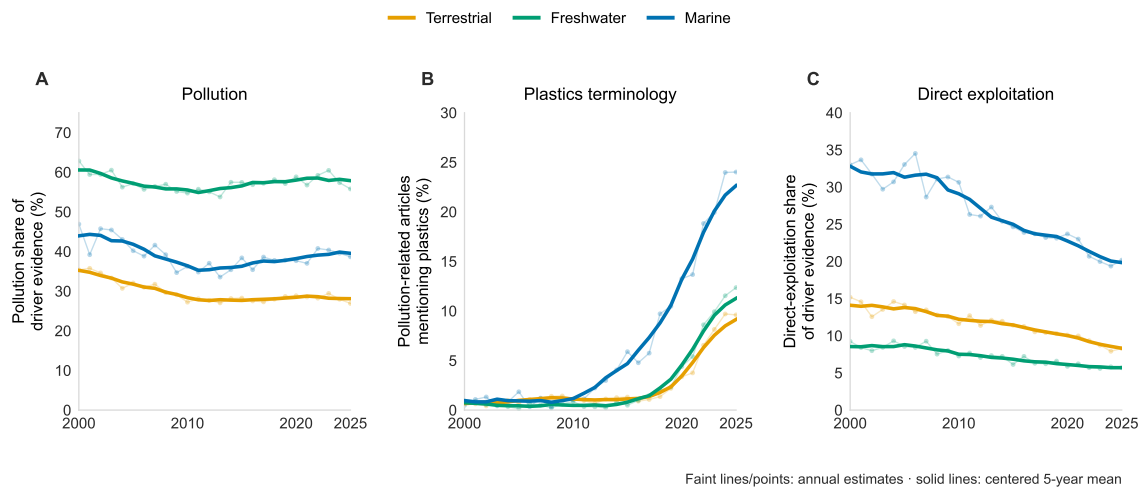
Supplementary Figure S1: Eligibility screening procedure: Each record is screened using four questions (Q1–Q4) applied to its title and abstract: Q1 biodiversity change (does the record report at least one form of biodiversity change, operationalized across genetic, species, community, and ecosystem change); Q2 direction of change (does the record identify a direction of biodiversity change, recorded as negative, positive, mixed, or unclear, with negative change, i.e. loss, as the primary analytical focus of the findings); Q3 direct anthropogenic driver (does the record mention or plausibly imply at least one driver, assessed using the five IPBES driver categories and the IUCN Threats Classification Scheme); and Q4 linkage (does the record report or plausibly imply a link between the identified driver and the biodiversity change). A record is classified as eligible only when all four questions return *Unclear* or *Yes*; a single *No* at any question yields an ineligible classification. Each question produces a structured label alongside its result. Full operationalization is provided in the screening prompt in the available code.



Supplementary Figure S2: **ROSES flow diagram:** Of 605,199 records meeting the pre-registered eligibility criteria, 253,081 reported a negative direction of change and constitute the evidence base for biodiversity loss analyzed in this study. As pre-registered in the protocol, records with a positive, mixed, or unclear direction of change (n = 352,118) were retained in the eligible set but excluded from synthesis, since biodiversity loss is the primary analytical focus of this map. A detailed breakdown of records excluded at the eligibility screening stage, by exclusion reason, is provided in Supplementary Figure S3.



Supplementary Figure S3: Overlap of screening criteria among ineligible records: Each column shows one exact combination of screening questions (Q1–Q4, see Supplementary Figure S1) answered No, ordered by record count. Most exclusions failed all four criteria simultaneously ($n = 1,081,249$, 59.8% of classified exclusions); a further 24.10% ($n = 435,966$) failed Q1, Q2, and Q4 jointly, and 12.03% ($n = 217,696$) failed Q3 and Q4 jointly. The remaining combinations each contributed between 0.01% and 2% of classified exclusions.



Supplementary Figure S4: **Research Attention to pollution and direct exploitation over time within each ecosystem realm, 2000–2025:** Faint lines and points show annual estimates; solid lines show centered five-year means. **(a)** Pollution as a share of driver attributions within each realm, which remained broadly stable across the period. **(b)** The share of pollution articles using plastic or marine-debris terminology, which rose from a negligible fraction to roughly a quarter of marine pollution articles and to around one in ten in the terrestrial and freshwater realms. **(c)** Direct exploitation as a share of driver attributions within each realm, which declined in all three realms and most steeply in marine evidence. Panels (a) and (c) share the fractional-attention definition used in Figure 3.

406 *Robustness of the income-composition contrast*

407 This section documents the analytical choices and robustness checks underlying the find-
408 ing that the documented-threat composition of the biodiversity-loss evidence differs across
409 national income groups. The unit of analysis is a scientific article reporting negative-
410 direction, observational evidence for a country of study; articles reporting more than one
411 broad IUCN threat category contribute fractionally, each of the k reported categories re-
412 ceiving weight $1/k$.

413 **Historical income matching:** Because a country's World Bank income classification
414 can change over time, each article was matched to the income group in force in its pub-
415 lication year rather than to the current classification. For complete publication years
416 2000–2025, this yielded 94,443 articles spanning 213 countries, and reclassified 11.0%
417 of article–country assignments relative to current income groups. The composition con-
418 trasts below are therefore computed on income groups as they stood at the time of study.

419 **Standardization for region and period:** Income groups differ in which world regions
420 and publication periods they draw on, so the raw contrasts could reflect the geography and
421 timing of research rather than income. To test this, we grouped the four income groups
422 into a lower-income tier (low and lower-middle) and a higher-income tier (upper-middle
423 and high), then directly standardized both tiers to a common distribution over World Bank
424 region and five-year publication period, using the 28 region–period strata present in both
425 tiers. The four principal contrasts persisted (Table S1): after joint standardization, lower-
426 income evidence remained more oriented toward agriculture and biological resource use,

427 and higher-income evidence toward biological invasions and climate change, with all four
 428 bootstrap intervals excluding zero. Pollution behaved differently: its small raw contrast
 429 fell to near zero under region adjustment, so we do not treat it as a stable income difference.

Threat category	Raw	Region-std.	Joint-std.	95% CI (joint)
Invasive & problematic species	6.2	5.3	5.2	4.4 to 6.0
Climate change & severe weather	3.2	3.9	3.0	1.1 to 3.9
Pollution	1.1	−0.1	1.5	0.2 to 2.8
Natural system modifications	3.3	0.8	1.2	0.5 to 1.8
Biological resource use	−7.0	−5.6	−5.4	−6.3 to −4.4
Agriculture & aquaculture	−6.5	−6.0	−7.0	−7.9 to −5.9

Supplementary Table S1: **Higher-income-minus-lower-income threat-share contrasts under standardization:** Differences in each broad IUCN threat category’s share of attributions between the higher- and lower-income tiers, in percentage points, shown raw, after region-only direct standardization, and after joint region-and-period standardization. Positive values indicate greater relative attention in higher-income evidence. Confidence intervals are bootstrap intervals for the joint-standardized contrast. Categories with contrasts near zero across all specifications are omitted.

430 **Sensitivity analyses:** The pattern was stable across alternative specifications. Using
 431 all negative-direction study designs rather than observational articles only, using current
 432 rather than historical income classification, shifting the fiscal-year boundary, including the
 433 partial 2026 year, and applying country-fractional weighting each preserved the direction
 434 of every contrast and the rank ordering of threats (Spearman rank correlation 0.99 to 1.00),
 435 with no single contrast shifting by more than 4.2 percentage points. These results describe
 436 documented research attention, not the ecological prevalence or severity of threats.

437 Supplementary methods

438 This section provides the methodological details underlying the methods section, covering
439 the literature search, validation datasets, and screening and coding results.

440 *Supplementary methods: Search*

441 The literature search was conducted on the WOS for publications from 2000 to April 2026,
442 with no language restrictions. The search string was designed to prioritize recall over pre-
443 cision; its five components are combined with the AND operator (see Supplementary Ta-
444 ble S2). Component (A) restricts retrieval to journal articles; component (B) limits results
445 to biodiversity-relevant WoS subject categories; component (C) requires language indi-
446 cating measured change, assessment, or monitoring; component (D) supplies five driver-
447 specific sub-strings, one per IPBES direct driver category, adopted from a prior rapid evi-
448 dence assessment [10], combined with OR; and component (E) applies the publication-year
449 filter. Gray literature was excluded because its processing at this corpus scale falls outside
450 the scope of the LLM pipeline, a recognized limitation of the search coverage.

451 We assessed search comprehensiveness against an *a priori* test list of 581 records
452 drawn from five published sources (three meta-analyses, an evidence assessment, and a
453 systematic map protocol) on direct anthropogenic drivers of biodiversity change [9, 10,
454 21–23] and the IPBES Global Assessment chapters on direct drivers (sections 2.1.12–
455 2.1.17) [24]. The list was assembled by extracting reference lists from each source and
456 excluding gray literature and records not indexed in WoS, yielding 581 records from an
457 initial 791 candidates; scientific publications cited by the original authors as evidence were
458 treated as eligible for this map. A comprehensiveness check conducted on 02 May 2026
459 found that the final search string retrieved 497, an overall recall of 85.54%, with source-
460 specific recall ranging from 71.43% to 97.32% (see Supplementary Table S3).

Component	Operator	What this block does	WoS field	Search string
(A) Publications	AND	Limit results to articles only.	DT	DT=(Article)
(B) Scientific disciplines	AND	Limits results to relevant Web of Science Categories to keep retrieval manageable.	WC	WC=("Biodiversity Conservation" OR "Ecology" OR "Environmental Sciences" OR "Environmental Studies" OR "Marine & Freshwater Biology" OR "Fisheries" OR "Oceanography" OR "Forestry" OR "Water Resources" OR "Agronomy" OR "Geosciences, Multidisciplinary" OR "Meteorology & Atmospheric Sciences" OR "Plant Sciences" OR "Zoology" OR "Remote Sensing" OR "Multidisciplinary Sciences")
(C) Change / assessment / monitoring terms	AND	Ensures retrieved records include language indicating measured change or assessment/monitoring (e.g., loss/change, impacts/pressures, trends/responses, risk, monitoring).	TS	TS=((loss* NEAR/3 (habitat* OR biodiversit* OR species OR population* OR "ecosystem service*" OR ecosystem* OR forest* OR coral* OR mangrove* OR reef* OR wetland* OR bird* OR mammal* OR plant*)) OR change* OR reference* OR indicator* OR impact* OR effect* OR pressure* OR trend* OR response* OR status OR pattern* OR assessment* OR monitor* OR risk* OR driver*)
(D) Direct drivers of anthropogenic biodiversity change (driver sub-strings combined with OR within TS)				
(D1) Land/sea use change	OR	Terms capturing land/sea use change and habitat modification.	TS	((("land use chang*" OR "land cover chang*" OR "land* system* chang*" OR "land* chang*" OR "land* degradation" OR "land* conversion" OR "land management") OR ((marine OR ocean OR sea*) AND use AND change*) OR ("habitat loss*" OR "habitat degradation*" OR "habitat chang*" OR "habitat conversion" OR "habitat modification*" OR "habitat fragmentation" OR "habitat management") OR (biofuel* OR biogas*) OR ((residential OR urban* OR cities OR housing OR commercial OR industrial OR tourist* OR recreation*) AND (extent OR area OR cover* OR proportion* OR percentage*) AND (develop* OR expand* OR increase*) OR (deforestation OR afforestation OR "wood plantation*" OR "forest* plantation*" OR "pulp* plantation*") OR ((aquaculture* OR agricultur* OR crop* OR "cultivated area*" OR farmland OR pasture) AND (develop* OR expand* OR increase* OR intensification OR extensification OR abandon* OR change*)) OR (livestock AND (farm* OR ranch* OR graz* OR densit*)) OR (energy AND (drill* OR min* OR quarry*)) OR (((transport* OR service) AND (network* OR corridor* OR infrastructure*)) OR road* OR rail* OR "ship* lane*" OR "flight path*") OR ("system* modification*" OR (dams OR (water AND (management OR use))))))
(D2) Direct Exploitation and Resource Extraction	OR	Terms capturing extraction/harvest/withdrawal of biological and abiotic resources, including fishing and logging.	TS	(((((natur* AND resource*) OR material* OR (biomass OR "fossil fuel*" OR ore* OR mineral*)) OR ((biological AND resource*) OR wildlife OR animal* OR plant* OR population*) OR water) AND ((overextraction OR overexploitation) OR (extraction OR exploitation OR withdraw*) OR depletion OR (hunting OR collect* OR gather* OR poaching OR "bush meat*")))) OR ((forest* OR wood OR timber*) AND (harvest* OR logging*)) OR (soil* AND (erosion OR acidification OR degradation)) OR (fisher* OR fishing OR (bycatch OR by-catch))
(D3) Climate change	OR	Terms capturing climate change and associated climatic stressors and impacts.	TS	("climat* change" OR "chang* climat*" OR ("increase* temperature*" OR "thermal alteration" OR "global warming") OR (chang* AND precipitation*) OR ((extreme OR severe) AND (weather OR temperature OR precipitation OR climat*)) OR heatwave* OR drought* OR storm* OR flood* OR (UV OR ultraviolet) OR "sea level rise" OR "soil salinization" OR ("habitat shift*" AND climate) OR ("habitat alteration" AND climate) OR ((consumptive OR human OR agricultural) AND "water stress"))
(D4) Pollution	OR	Terms capturing pollution and contaminants across environmental media (soil/water/air).	TS	((pollut* OR contamin* OR emission* OR spill-over* OR disposal* OR deposition* OR load* OR dump* OR discharge*) OR ((noise OR light OR gas* OR particle* OR particulate* OR nitrogen OR sulphur* OR phosphor* OR waste* OR garbage OR sewage OR pesticide* OR fertilizer* OR nutrient* OR "oil spill*" OR metal OR salinization OR salinisation OR acidification OR solid OR plastic OR mercury OR nutrient* OR pollutant*) AND (soil OR water OR ocean OR marine OR atmosphere OR air)) OR ((industrial OR military OR agricultur* OR forest*) AND effluent*))
(D5) Invasive alien species	OR	Terms capturing invasive alien species introductions and biological invasions.	TS	((((invasive OR invasion) AND (alien OR exotic OR non-native OR pest* OR disease) AND species) OR "biological invasion*" OR (introduc* AND (species OR material))))
(E) Publication year	AND	Restricts results to the selected publication year range.	PY	PY=2000-2026

Supplementary Table S2: **Search string components used in Web of Science:** Components (A)–(E) are combined using AND. Within TS, the five direct driver sub-strings (D1–D5) from [10] are combined using OR. Component (E) applies the WoS publication-year filter PY=2000–2026; because WoS publication-year filtering does not express a day/month cutoff, records with publication-date metadata after 30 April 2026 were excluded at the point of export.

Source	DOI	Test list (<i>n</i>)	Retrieved	Recall (%)
Beckmann et al. [21]	10.1111/gcb.14606	105	75	71.43
Benítez-López et al. [22]	10.1126/science.aaj1891	60	52	86.67
Balvanera et al. (IPBES 2.1.12–2.1.17) [24]	10.5281/zenodo.3831673	119	96	80.67
Jaureguiberry et al. [10]	10.1126/sciadv.abm9982	139	128	92.09
Moullec et al. [9]	10.1186/s13750-021-00234-y	46	37	80.43
Urban [23]	10.1126/science.aaa4984	112	109	97.32
Total	–	581	497	85.54

Supplementary Table S3: **Comprehensiveness of the search against the *a priori* test list:** For each source, the test list count is the number of eligible records indexed in WoS, and recall is the percentage retrieved by the final search string. The test list is the union of all sources ($n = 581$), against which overall recall is assessed.

461 *Supplementary methods: Consistency-check datasets*

462 This map deviates from CEE guidelines by using LLMs rather than human reviewers and
 463 by screening from abstracts rather than full text; its outputs are therefore validated against
 464 expert full-text labels: labels assigned by the original authors of the reference sources
 465 through full-text reading. Four datasets support this validation: two drawn from previously
 466 published evidence syntheses containing those expert labels, and two manually annotated
 467 by the authors of this map from titles and abstracts only. Consistency Check 1 (CC1)
 468 compares author abstract-level labels against expert full-text labels to establish a human-
 469 performance baseline; Consistency Check 2 (CC2) evaluates LLM pipeline labels against
 470 the expert labels. Supplementary Table S4 summarizes the four datasets; Supplementary
 471 Table S5 lists the secondary sources from which they were compiled. All datasets are
 472 shared in the available data for transparency and reproducibility.

Dataset	Size	Details	Role
Reference Screening Dataset	$n = 2,418$	Records included in three meta-analyses, a rapid evidence assessment, and the Red List of Ecosystems Database.	CC2: LLM labels evaluated against expert full-text labels to assess screening recall
Annotated Screening Dataset	$n = 500$	Author-assigned screening labels based on abstracts only, samples drawn from the Reference Screening Dataset and search query results. No full-text access allowed during annotation.	CC1A: author abstract-level labels compared against expert full-text labels. CC1B: LLM labels compared against author-annotated labels.
Reference Coding Dataset	$n = 7,844$	Records included in three meta-analyses, two systematic reviews, two systematic maps, a rapid evidence assessment, and one preprint, carrying expert full-text labels across applicable label sets L1–L6.	CC2: LLM labels evaluated against expert full-text labels across all six label sets (L1–L6)
Annotated Coding Dataset	$n = 600$ (100 per label set)	Author-assigned coding labels based on abstracts only, samples drawn from the Reference Coding Dataset. No full-text access allowed during annotation.	CC1: labels assigned from abstracts compared against expert full-text labels across all six label sets (L1–L6)

Supplementary Table S4: Overview of the four validation datasets supporting the screening and data-coding consistency checks.

473 **Reference Screening Dataset** ($n = 2,418$): Compiled from three meta-analyses, a rapid
474 evidence assessment, and the Red List of Ecosystems Database, the Reference Screening
475 Dataset contains 2,418 records. In each source, the original authors read records in full and
476 retained them as eligible; these full-text inclusion decisions are treated as ground-truth la-
477 bels, with the alignment between source inclusion criteria and the eligibility criteria of this
478 map treated as a working assumption. The dataset is partitioned into train, development,
479 and test splits (60/20/20, stratified by source) and supports CC2.

480 **Annotated Screening Dataset** ($n = 500$): The Annotated Screening Dataset comprises
481 500 records split into two sub-samples and independently annotated by the authors from
482 titles and abstracts only. The *reference dataset sample* ($n = 100$) is drawn from the Ref-
483 erence Screening Dataset and supports CC1A: author abstract-level labels are compared
484 against expert full-text labels to establish a human recall baseline. The *search sample*
485 ($n = 400$) is drawn directly from search-query retrieval results and annotated by the au-
486 thors from titles and abstracts only; it supports CC1B: a blind comparison of LLM screen-
487 ing decisions against author-annotated labels on search results.

488 **Reference Coding Dataset** ($n = 7,844$): The Reference Coding Dataset was compiled
489 from three meta-analyses, two systematic reviews, two systematic maps, a rapid evidence
490 assessment, and one preprint. Labels were produced by the original authors of each source
491 through full-text coding as part of their respective reviews; these author-assigned labels
492 are treated here as expert ground truth against which LLM coding outputs are evaluated
493 across applicable labels within each of the six label sets (L1–L6).

494 **Annotated Coding Dataset** ($n = 600$): The Annotated Coding Dataset comprises of
495 100 records per label set drawn from the Reference Coding Dataset, with original expert
496 labels stripped before the authors independently re-labeled each record from titles and
497 abstracts only. The dataset supports CC1: author abstract-level labels are compared against
498 expert full-text labels per label set to establish a human-performance baseline.

499 **Source inventory:** The datasets and test list draw on 14 previously published reviews,
500 maps, and databases (one pre-print); Supplementary Table S5 lists each source with its
501 contributions across the test list and validation datasets. For each source, raw bibliographic
502 and label data were standardized through three steps: records were matched to their WOS
503 bibliographic entries; label values were harmonized to the controlled vocabulary used in
504 this map where source terminologies differed; and records were restructured so that each
505 row corresponds to one record-label-set pair. Full provenance documentation is shared in
506 the available data.

Source	Topic / scope	Test list (<i>n</i>)	Ref. Screening (<i>n</i>)	Ref. Coding (label[<i>n</i>])
Beckmann et al. [21]	Land-use intensification and species richness	105	-	-
Balvanera et al. [24]	IPBES Global Assessment Ch. 2.1, sections 2.1.12–2.1.17	119	-	-
Jaureguiberry et al. [10]	Direct drivers of biodiversity loss	139	128	region[128], realm[128], driver[128]
Moullec et al. [9]	Anthropogenic drivers in the North Sea	46	-	driver[1,978], study_design[2,007]
Benítez-López et al. [22]	Hunting impacts on tropical mammals and birds	60	-	region[48], country[48], realm[48], biome[48], kingdom[49], phylum[49], class[49], order[49], species[49]
Urban [23]	Extinction risk from climate change	112	-	region[103], sub-region[42], country[14], kingdom[100], phylum[37], class[32]
Keck et al. [26]	Global human impact on biodiversity	-	1,711	study_design[2,122], realm[2,122], driver[2,122]
Shaw et al. [25]	Genetic diversity change	-	418	-
Murphy and Romanuk [27]	Community change and species richness	-	142	-
Red List of Ecosystems Database [33]	Ecosystem risk assessments	-	31	-
Wright et al. [34]	IUCN threats and status (mustelids)	-	-	region[608], sub-region[534], country[513], threats_10[563], threats_11[539], kingdom[601], phylum[601], class[601], order[601], genus[601], species[601]
Lowry et al. [35]	Biological invasions	-	-	region[1,483], sub-region[1,483], country[1,477], realm[1,252]
Ridley et al. [36]	Threats to species globally	-	-	study_design[693], realm[693]
Pulido-Chadid et al. [37]	Effectiveness of protected areas	-	-	threats_10[73], threats_11[52]

Supplementary Table S5: Source inventory of the 14 published sources (one pre-print) used across the test list, Reference Screening Dataset, and Reference Coding Dataset.

507 *Supplementary methods: Screening*

508 Screening evaluated the corpus of literature retrieved from WoS against the eligibility
509 criteria pre-registered in the protocol, aiming to identify records reporting an aspect of
510 biodiversity change linked to a direct anthropogenic driver (see Supplementary Figure S1).

511 To assess the reliability of this fully automated procedure, screening decisions were
512 validated through two consistency checks against expert labels drawn from previously
513 published evidence syntheses. Records were labeled independently by an author from ti-
514 tles and abstracts and by the LLM, with reliability assessed primarily via recall, alongside
515 precision and F1 where relevant. This section presents the ROSES flow diagram summa-
516 rizing record counts from retrieval to final eligible set, followed by the validation results.

517 **ROSES flow diagram:** Supplementary Figure S2 presents the ROSES flow diagram [38]
518 documenting record counts across the screening pipeline. Of 2,414,191 records entering
519 eligibility screening, 605,199 (25.1%) were retained as eligible. Exclusions were con-
520 centrated in a small number of criterion combinations (see Supplementary Figure S3).
521 Most exclusions (59.8%) failed all four criteria simultaneously. A further 24.10% failed
522 Q1 (biodiversity change), Q2 (direction of change), and Q4 (driver–biodiversity linkage)
523 jointly, while reporting an anthropogenic driver (Q3). Together, these two combinations
524 accounted for 83.90% of all classified exclusions.

525 As pre-registered in the Supplementary Text S1 of the protocol, eligibility does not
526 require a negative direction of change: records reporting positive, mixed, or unclear di-
527 rection were retained as eligible but excluded from synthesis, since biodiversity loss is the
528 primary analytical focus of this map. Of the 352,118 eligible records not analyzed in this

study, 141,267 (23.34% of eligible records) reported mixed direction, 112,684 (18.62%) reported positive direction, and 98,167 (16.22%) reported unclear direction (Supplementary Table S6).

Direction of change	Records	% of eligible
Negative (biodiversity loss)	253,081	41.82%
Mixed	141,267	23.34%
Positive	112,684	18.62%
Unclear	98,167	16.22%

Supplementary Table S6: Distribution of direction of change (Q2) across the 605,199 eligible records. Negative direction constitutes the biodiversity loss dataset analyzed in this study (Supplementary Figure S2); the remaining categories are retained in the eligible set but excluded from synthesis.

Screening validation results: The screening prompt operationalizes the four-question procedure as a zero-shot task: GPT-5-nano receives the title and abstract of each record alongside step-by-step guidelines for Q1–Q4, covering biodiversity change, direction of change, direct anthropogenic driver, and driver–biodiversity linkage, returning a structured Yes, Unclear, or No result and a content label for each step. To prioritize recall, the prompt instructs the model to prefer cautious inclusion over strict exclusion, using Unclear rather than No when evidence is weak.

We first established how well a human reviewer can screen records from titles and abstracts alone, without full-text access. Comparing author abstract-level labels against expert full-text labels, this human baseline achieved a recall of 83% (CC1A). We then evaluated whether the LLM, working under the same abstract-only constraint, could match this baseline: reasoning effort was varied on the development split, the best-performing

configuration selected and confirmed on the training split, and evaluated on a held-out test split (CC2). The LLM achieved a recall of 90.45% on the held-out test split, exceeding the human baseline (see Supplementary Table S7).

Check	Comparison	Split [<i>n</i>]	Reasoning effort	Recall (%)
CC1A	Author abstract-level labels vs. expert full-text labels	[100]	–	83.00
CC2	LLM labels vs. expert full-text labels	Dev [484]	Low	88.42
			Medium	90.28
			High	90.28
		Train [1,452]	Medium	89.66
		Test [482]	Medium	90.45

Supplementary Table S7: **Screening validation results (CC1A and CC2)**. Recall is the primary metric, prioritized to minimize false exclusions. CC1A establishes a human recall baseline by comparing author abstract-level labels against expert full-text labels. For CC2, reasoning effort was varied on the development split; the medium configuration was selected and confirmed on the training and held-out test splits.

CC2 results, however, are based on the Reference Screening Dataset, which is drawn from prior published reviews where every record was already deemed eligible by the original authors. A reasonable concern is whether the LLM behavior changes when applied to real-world search results rather than this curated benchmark. To check this, we compared LLM and human screening decisions on a separate sample of 400 records drawn directly from search retrieval (CC1B), treating human labels as ground truth. This comparison measures agreement between the two abstract-only labelers, providing a complementary signal to CC2. The LLM achieved a recall of 90.1% against human labels (precision 64.5%, F1 75.2%; see Supplementary Table S8), closely consistent with CC2 test-split recall of 90.45%.

The lower precision reflects the screening design’s recall-prioritizing rule of retaining Unclear cases as eligible: a record was treated as eligible unless one of the four

559 screening stages was explicitly coded as No (0), so uncertain stage outputs (-1) remained
560 eligible by default. Of the 60 false positives in CC1B, 17 (28.3%) contained at least one
561 uncertain stage decision. Reclassifying these conservatively as ineligible increased pre-
562 cision from 64.5% to 69.3% but reduced recall from 90.1% to 80.2% (F1 74.3%): the
563 high-recall operating point partly reflects permissive treatment of uncertainty, not the de-
564 sign rule alone.

565 The remaining 43 false positives (71.7%) were confident errors. The corresponding
566 human exclusion reasons showed these errors concentrated near the conceptual bound-
567 ary of the screening definition: the two largest categories were records lacking a clear
568 biodiversity-change endpoint (13/43, 30.2%) and records where the anthropogenic driver
569 or its linkage to biodiversity change was absent (12/43, 27.9%). Most residual precision
570 loss therefore traced not to irrelevant retrievals but to abstracts that were environmentally
571 or biodiversity-adjacent while lacking an explicit instance of biodiversity change, an an-
572 thropogenic driver, or a clear link between the two. Together, these results show that
573 the LLM’s abstract-level screening behavior remains consistent across both the curated
574 reference dataset and uncured, real-world search results, with recall, the primary met-
575 ric guiding the screening design, showing no evidence of degradation beyond the curated
576 benchmark; precision proved more sensitive to how borderline and uncertain cases were
577 resolved.

	Human (truth)	
	Eligible	Ineligible
LLM Eligible	109	60
LLM Ineligible	12	219

Supplementary Table S8: **Search-sample agreement matrix (CC1B)**: LLM and human abstract-only screening decisions on 400 records drawn directly from search retrieval results, treating human labels as ground truth and "Eligible" as positive class.

578 **Screening output fields:** Each bibliographic record processed through the screening
579 pipeline carries a structured output comprising three tiers, regardless of eligibility deci-
580 sion. First, provenance fields support full transparency and reproducibility of each screen-
581 ing decision. Second, per-question structured labels capture the reasoning behind each de-
582 cision: biodiversity change type(s) for Q1, direction of change for Q2, identified driver(s)
583 for Q3, and linkage type for Q4. These labels provide richer evidence than the binary
584 decision alone. Third, a binary eligibility decision (`pred_screening`) records whether
585 the record passed the four-question procedure; 605, 199 records were retained as eligible.
586 The complete output schema is presented in Supplementary Table S9.

Field	Type	Description
<i>Metadata</i>		
_surrogate_key	String	Primary key of the record processed in the screening and coding stage.
_source_run	String	Identifier for the processing partition in which the record was screened.
model	String	Name of the LLM used to generate the screening output.
created_at	Datetime	Timestamp indicating when the record was processed.
raw_output	String	Original LLM response before parsing into structured fields.
<i>Screening outputs Q1–Q4</i>		
q1_r	Integer	Q1 result: biodiversity change reported? 1 = Yes, –1 = Unclear, 0 = No.
q1_label	List[String]	Biodiversity change type(s): genetic change, species change, community change, or ecosystem change.
q2_r	Integer	Q2 result: direction identifiable? 1 = Yes, –1 = Unclear, 0 = No.
q2_label	List[String]	Direction label(s): negative, positive, mixed, or unclear.
q3_r	Integer	Q3 result: direct anthropogenic driver mentioned or plausibly implied? 1 = Yes, –1 = Unclear, 0 = No.
q3_label	List[String]	Driver or threat name(s) identified during screening; these support eligibility screening and are not the final coding outputs.
q4_r	Integer	Q4 result: driver–biodiversity link reported or plausibly implied? 1 = Yes, –1 = Unclear, 0 = No.
q4_label	List[String]	Linkage type(s) describing how the driver is linked to the reported biodiversity change.
<i>Eligibility decision</i>		
pred_screening	List[String]	Final eligibility decision derived from Q1–Q4: ["ELIGIBLE"] or ["INELIGIBLE"].

Supplementary Table S9: **Output fields recorded for each record during screening:** Screening fields (Q1–Q4) record the decision result and structured label(s) generated for each eligibility question.

587 *Supplementary methods: Coding*

588 All records classified as eligible during screening entered data coding across six label sets
589 (L1–L6), together capturing the evidence chain from anthropogenic driver to biodiversity
590 impact. Prompts, configurations, and the full output schema for each label set are shared in
591 the available code; validation status per label is summarized in Supplementary Table S10.
592 Of the 23 labels produced, 14 are validated through full CC1 and CC2 consistency checks
593 and form the primary basis for the results below; the remaining 11 are weak, for reasons
594 detailed later in this section. This section presents CC1 coding validation, followed by
595 CC2 coding validation, and closes with an account of the weak labels.

Label set	Label	Type	Description	Is Validated	Is Weak
<i>L1: IPBES Direct Drivers</i>					
L1	driver	List[String]	IPBES direct driver category or categories assigned to the record.	Yes	-
<i>L2: IUCN Threats</i>					
L2	pred_threat_l10	List[String]	Broadest IUCN Threats Classification level assigned to the record.	Yes	-
L2	pred_threat_l11	List[String]	Intermediate IUCN Threats Classification level assigned to the record.	Yes	Yes*
L2	pred_threat_l12	List[String]	Most specific IUCN Threats Classification level assigned to the record.	-	Yes
<i>L3: Geographic Scope</i>					
L3	pred_regions	List[String]	IPBES region or regions of study focus.	Yes	-
L3	pred_subregions	List[String]	IPBES sub-region or sub-regions of study focus.	Yes	-
L3	pred_countries	List[String]	Countries of study focus, recorded using ISO 3166-1 alpha-3 codes.	Yes	-
L3	locales	List[String]	Free-text names of sub-national locations mentioned in the title or abstract, such as parks, reserves, cities, or mountain ranges.	-	Yes
L3	locale_coordinates	List[List[Float]]	[latitude, longitude] pairs in decimal degrees (WGS84, EPSG:4326); resolved from explicit coordinates in the title or abstract, or inferred from the locale name if absent.	-	Yes
<i>L4: Ecosystem Typology</i>					
L4	pred_realm	List[String]	Global Ecosystem Typology realm or realms of study focus.	Yes	-
L4	pred_biome	List[String]	Global Ecosystem Typology biome or biomes of study focus.	-	Yes*
L4	pred_efg	List[String]	Global Ecosystem Typology Ecosystem Functional Group or groups of study focus.	-	Yes*
<i>L5: Study Attributes</i>					
L5	pred_study_design	String	Study design category: observational, experimental, modelling, or unclear.	Yes	-
L5	pred_methods_data_collection	List[String]	Data collection method or methods reported in the title or abstract, where applicable.	-	Yes
L5	pred_methods_analysis	List[String]	Analysis method or methods reported in the title or abstract, where applicable.	-	Yes
L5	pred_has_comparison	Boolean	True if the record explicitly states or strongly implies a comparator, such as pre/post, control versus exposed sites, or reference sites; otherwise false.	-	Yes
L5	pred_comparison_types	List[String]	Type or types of comparison identified for the record, where applicable.	-	Yes
<i>L6: Taxonomy</i>					
L6	kingdom	List[String]	Kingdom(s) resolved through the GBIF Backbone Taxonomy.	Yes	-
L6	phylum	List[String]	Phylum/phyla resolved through the GBIF Backbone Taxonomy.	Yes	-
L6	class	List[String]	Class(es) resolved through the GBIF Backbone Taxonomy.	Yes	-
L6	order	List[String]	Order(s) resolved through the GBIF Backbone Taxonomy.	Yes	-
L6	genus	List[String]	Genus/genera resolved through the GBIF Backbone Taxonomy.	Yes	-
L6	species	List[String]	Species resolved through the GBIF Backbone Taxonomy.	Yes	Yes*

Supplementary Table S10: Coding output fields and validation status: Of the 23 labels produced during data coding, 14 are designated *validated*: evaluated through the full CC1 and CC2 consistency checks and formed the primary basis for the validation metrics and the results. 11 labels are designated *weak*, for one of three reasons. 7 weak labels (unmarked) were not validated from the outset, as no standardized labeling schema comparable to the classification schemes used elsewhere exists for these fields, or, for `pred_threat_l12`, because the label applies only to a narrow subset of threat branches. 2 weak labels (marked with an asterisk) were evaluated through CC1 and CC2 but showed weak performance in these checks: `pred_threat_l11` and `species`. Two further weak labels (marked with an asterisk) were not evaluated through CC1 and CC2 due to the absence of available expert-labeled data, and were instead manually spot-checked by the authors against LLM predictions: `pred_biome` and `pred_efg`. All unvalidated and weakly validated labels are nonetheless included in the map for completeness. A hyphen (-) indicates that neither weak nor validation reason applies.

CC1 coding validation: CC1 established a human-performance baseline for data coding, testing whether the label sets are recoverable from titles and abstracts alone; the same information constraint under which the LLM operates. For each label set, the author assigned labels to 100 records drawn from the Annotated Coding Dataset using titles and abstracts only, and compared these abstract-level labels against expert full-text labels from the Reference Coding Dataset; records where predictions defaulted to masked values (such as `Unclear` or `Not Applicable`) or fell outside the label schema were excluded from evaluation, so the effective n per label set varies (see Supplementary Table S11). All six label sets involve multi-label classification, where a record may carry multiple values per label set; this structurally depresses agreement metrics relative to single-label settings.

CC1 achieved a macro-average weighted F1 of 84.24% and Cohen’s κ of 0.75 across label sets (see Supplementary Table S11). Performance varies across label sets, reflecting differences in task complexity: the IPBES driver task (`L1`) involves only five possible categories, while IUCN threats (`L2`) spans 12 categories at level 0 alone and grows considerably as the hierarchy deepens. This difference is evident in the results: `threats_l0` achieved an F1 of 75.06%, while `threats_l1`, which requires identifying the specific mechanism of a threat, fell to 57.62%. For instance, a record covering threats from residential and commercial development may be correctly identified at `threats_l0`, but it is often not possible to determine whether the origin is urban, commercial, or recreational expansion from an abstract alone. For `L6` (Taxonomy), lower-rank labels such as genus (F1 95.31%) and species (F1 76.65%) are the meaningful performance indicators; kingdom and phylum scores (99.53% and 100% respectively) reflect data composition: the validation sample for phylum contained a single dominant class (*Chordata*) and should be

619 interpreted conservatively. Taken together, CC1 establishes that validated labels are re-
620 coverable from titles and abstracts at the performance levels shown, providing the human
621 baseline against which LLM performance is assessed in CC2.

Label set	Label	n	Precision (%)	Recall (%)	F1 (%)	κ
L1 – IPBES Direct Drivers	driver	100	86.60	79.44	81.94	0.76
L2 – IUCN Threats	threats_l0	86	82.22	70.15	75.06	0.67
	threats_l1	49	72.51	51.47	57.62	0.53
L3 – Geographic Scope	region	100	95.02	90.20	92.38	0.89
	sub-region	91	96.35	91.40	93.33	0.90
	country	90	95.10	81.91	87.39	0.84
L4 – Ecosystem Typology	realm	100	100	93.07	96.14	0.93
L5 – Study Attributes	study_design	100	82.31	76.47	76.76	0.52
L6 – Taxonomy	kingdom	100	99.12	100	99.53	0.99
	phylum	90	100	100	100	-
	class	90	94.15	95.56	94.81	0.63
	order	86	92.65	93.55	93.06	0.80
	genus	78	96.72	94.25	95.31	0.94
	species	85	78.48	76.03	76.65	0.76

Supplementary Table S11: **CC1 coding validation results:** Across label sets, CC1 achieved an overall macro-average weighted precision of 89.21%, weighted recall of 81.81%, weighted F1 of 84.24%, and weighted Cohen’s κ of 0.75. For this summary, weighted metrics were first macro-averaged within each label set and then macro-averaged across label sets. All reported precision, recall, and F1 values are weighted metrics; n denotes records valid for scoring after exclusion of masked or out-of-schema labels, such as unclear cases or instances where no ground-truth label was available. The *phylum* results should be interpreted with caution because the validation data contained only one observed class (*Chordata*); accordingly, Cohen’s κ is not applicable.

622 **CC2 coding validation:** CC2 evaluated LLM coding labels against expert full-text la-
623 bels across six label sets, providing the ground-truth accuracy check complementary to
624 CC1 human baseline. Prompts were iteratively developed on the training split and reason-
625 ing effort configurations compared on the development split; the best-performing config-
626 uration per label set was then evaluated on the held-out test split.

627 On the held-out test split, CC2 achieved a macro-average weighted F1 of 83.54% and
628 Cohen’s κ of 0.72 across label sets. As with CC1, all label sets involve multi-label clas-
629 sification, which structurally depresses agreement metrics relative to single-label settings
630 (see Supplementary Table S12).

631 Performance patterns mirror those observed in CC1 and reflect the same underlying
632 constraints. For L2 (IUCN Threats), `threats_l1` (F1 57.52%) remains considerably
633 lower than `threats_l0` (F1 68.43%); as in CC1, this reflects the specificity of sub-level
634 threat classification rather than a model limitation: the abstract-level text frequently does
635 not contain sufficient detail to distinguish. For L6 (Taxonomy), species-level performance
636 (F1 57.95%) falls notably below the CC1 human baseline (F1 76.65%). In CC1, hu-
637 man annotators could leverage contextual cues, sub-species disambiguation, and manual
638 taxonomy lookups; the LLM resolved species identity from text alone, which introduces
639 conflation between adjacent taxa. Importantly, this does not propagate upward through the
640 hierarchy: even where a species prediction is adjacent rather than exact: for instance, a dif-
641 ferent species within the same genus are resolved via GBIF Backbone Taxonomy lookup
642 and remain unaffected, as confirmed by the stronger genus performance (F1 93.88%) rela-
643 tive to species. Overall, CC2 broadly approaches CC1 performance across validated label
644 sets, indicating that LLM coding from titles and abstracts is a reliable approach to large-
645 scale systematic mapping.

Split	Metric	L1 driver	L2 threats I0	L2 threats I1	L3 region	L3 sub-region	L3 country	L4 realm	L5 study design	L6 kingdom	L6 phylum	L6 class	L6 order	L6 genus	L6 species
	Selected config <i>n</i>	Low 841	Medium 111	High 93	Low 471	Low 411	Low 411	High 843	Low 952	High 148	High 134	High 126	High 120	High 109	High 126
Dev Low	Precision	81.75	80.57	63.68	92.22	84.32	94.23	94.58	86.74	98.47	100	88.47	91.85	96.03	75.68
	Recall	88.96	69.88	56.12	89.60	89.50	75.53	85.31	82.89	95.97	98.51	90.62	88.15	96.55	69.23
	F1	85.06	73.43	56.42	90.53	86.31	82.57	89.49	84.30	97.19	99.25	88.82	89.71	96.02	70.35
	Cohen's κ	0.79	0.66	0.51	0.85	0.80	0.76	0.80	0.66	0.80	-	0.42	0.86	0.95	0.69
Dev Medium	Precision	80.92	82.31	65.27	90.23	83.83	94.69	93.93	86.32	98.21	100	87.70	91.91	95.97	74.56
	Recall	87.43	70.56	56.49	87.53	89.98	74.12	86.84	82.06	94.63	97.01	88.37	86.76	95.69	69.74
	F1	83.79	73.67	57.58	88.54	86.18	82.15	90.02	83.63	96.34	98.48	87.35	89.09	95.51	70.95
	Cohen's κ	0.78	0.66	0.51	0.82	0.80	0.75	0.81	0.65	0.75	-	0.29	0.77	0.94	0.70
Dev High	Precision	80.09	79.70	64.94	91.33	83.59	94.37	94.57	86.10	98.66	100	87.73	91.11	96.77	74.32
	Recall	86.67	70.17	58.97	89.19	91.41	74.35	86.84	81.34	97.32	98.50	89.84	88.15	97.39	70.26
	F1	82.97	72.84	58.60	89.74	86.65	82.15	90.44	83.06	97.98	99.24	88.06	89.43	96.85	71.22
	Cohen's κ	0.76	0.65	0.53	0.84	0.81	0.75	0.82	0.64	0.86	-	0.33	0.83	0.96	0.70
Train	<i>n</i>	2515	325	295	1410	1227	1223	2523	2856	441	396	384	366	343	384
	Precision	80.89	77.76	69.63	92.25	86.04	92.20	96.00	88.36	98.51	100	90.47	90.93	93.92	61.27
	Recall	87.58	65.77	54.32	90.08	91.28	77.52	89.13	84.99	96.16	97.73	91.49	88.75	92.43	53.42
	F1	84.01	70.84	59.09	90.76	88.13	83.57	92.30	86.37	97.31	98.85	90.57	89.36	92.67	55.03
	Cohen's κ	0.78	0.63	0.54	0.86	0.83	0.78	0.85	0.70	0.76	-	0.30	0.75	0.91	0.54
Test	<i>n</i>	837	111	97	469	404	402	839	950	146	131	126	119	113	125
	Precision	81.02	72.99	63.58	91.76	86.99	91.05	95.85	86.94	100	100	94.46	95.96	92.27	59.77
	Recall	87.05	65.73	56.13	90.02	91.21	77.08	89.73	82.97	97.96	99.24	96.09	93.55	95.76	58.29
	F1	83.85	68.43	57.52	90.38	88.41	82.97	92.61	84.60	98.90	99.62	94.95	94.54	93.88	57.95
	Cohen's κ	0.78	0.58	0.51	0.85	0.83	0.77	0.85	0.67	0.93	-	0.25	0.63	0.93	0.57

Supplementary Table S12: CC2 coding validation results: On the held-out test set, CC2 achieved an overall macro-average weighted precision of 85.41%, weighted recall of 82.82%, weighted F1 of 83.54%, and weighted Cohen's κ of 0.72. For this summary, weighted metrics were first averaged within each label set and then macro-averaged across label sets. All reported precision, recall, and F1 values are weighted metrics; *n* denotes records valid for scoring after exclusion of masked or out-of-schema labels, such as unclear cases or instances where no ground-truth label was available. The *phylum* results should be interpreted with caution because the validation data contained only one observed class (*Chordata*); accordingly, performance may be overstated and Cohen's κ is not applicable, reported as “-”.

646 **Weak labels:** As summarized in Supplementary Table S10, 11 of the 23 coding labels
647 are designated *weak*, for one of three reasons. Two, `pred_threat_l1` and `species`,
648 were evaluated through the full CC1 and CC2 consistency checks reported above and
649 are weak based on those results. Two further labels, `pred_biome` and `pred_efg`,
650 lacked expert-labeled ground truth and were instead audited manually. The remaining
651 seven lacked expert-labeled data and a standardized schema against which to validate;
652 `pred_threat_l2` is a partial exception, as its most specific IUCN level is structurally
653 absent from roughly half the truth set’s hierarchy branches, making validation largely in-
654 applicable rather than merely unattempted.

655 `pred_biome` and `pred_efg` were audited by comparing the authors’ manual read-
656 ing of titles and abstracts against LLM predictions for a sample of 50 records each.
657 `pred_biome` achieved a weighted precision of 94.11%, recall of 86.27%, F1 of 90.01%,
658 and Cohen’s κ of 0.65; `pred_efg` achieved a weighted precision of 83.92%, recall of
659 74.50%, F1 of 76.75%, and Cohen’s κ of 0.63. These results are reasonably strong de-
660 spite the absence of a formal validation cycle, likely because realm, biome, and ecosystem
661 functional group are often directly inferable from the ecological description given in a title
662 or abstract.

663 *Supplementary methods: Reporting checklist*

664 A reporting checklist [20] for this systematic map is presented in Extended Data Table 1,
665 documenting the transparency, reproducibility, and rationale behind majority of LLM us-
666 age decision in the pipeline. To further support reproducibility, the available code includes
667 a representative subset of LLM inputs, raw model outputs, and evaluation results in their
668 original form.

Extended Data Table 1: **Reporting checklist for LLM use:** The GUIDE-LLM checklist [26] is completed here to document the transparency and reproducibility of LLM use in this systematic map.

Item	Answer
Scope of LLM use	
Item A.1: LLMs were used in this project for: Explanation: Briefly describe how and for what purposes LLMs were used in the study. This may include one or multiple stages of the research workflow. The following examples illustrate common use cases: • Research design (e.g., hypothesis generation, literature search, or creating surveys/stimuli). • Data processing (e.g., transcription, translation, data extraction, or data cleaning). • Analysis (e.g., data labeling, summarization, pattern detection, statistical analysis, or code generation). • LLM as research object (e.g., studying LLM behavior, benchmarking LLMs, or bias assessment of LLMs). • Participant-facing settings (e.g., LLM used as an intervention, studying human interactions with LLM chatbots). • Communication (e.g., paper writing, editing, or reviewing). Depending on the specific use case described here, different checklist items may later be relevant.	LLM use involved eligibility screening of bibliographic records (title and abstract) using a four-question procedure (Q1–Q4) and data coding of eligible records across six label sets (L1–L6) covering anthropogenic drivers, threat classification, geographic scope, ecosystem typology, study attributes, and taxonomy.
Item A.2: Degree of automation (human-in-the-loop vs. fully automated): Explanation: Indicate how much human oversight was involved. Specify whether each output was reviewed, edited, or approved by a person, or whether outputs were used automatically without supervision. Specify who provided oversight (e.g., student assistant, expert, PI).	Primarily automated. All eligibility screening and data coding decisions on individual records were made by the model, with no per-record human review during full-corpus application. Human involvement occurred at defined stages: prompt design and iterative refinement on training and development splits prior to freezing, construction of validation datasets (CC1/CC2, covering over 8,000 human-annotated coding records and approximately 2,900 screening records), evaluation of pipeline performance on a held-out test split, and a post-hoc audit of a sample of screened and coded records conducted by the authors after full-corpus application.
Model/system details	

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Reporting checklist for LLM use (continued).

Item	Answer
Item B.1: Model name, including provider, model size, exact version/ID, date of access, and source link (if possible): Explanation: Report the exact model names (including provider, version, and date accessed). Avoid generic labels like “ChatGPT” or “GPT-4”; instead, use detailed model names. If multiple models were tested, name them and briefly explain which one was used in the final study and why.	GPT-5-nano (version 2025-08-07), provided by OpenAI. Accessed via the OpenAI API. The model and its version were frozen for selection prior to full-corpus application.
Item B.2: Model access (e.g., API, web interface, local) and context mode (e.g., chat mode or separate calls): Explanation: Note how you accessed the models and whether you used LLMs in chat mode (ongoing conversation) or stateless mode (separate prompts). Mention the exact API name and version, since different access modes may influence responses.	Access via the OpenAI Responses API. Each record was processed via stateless API calls with no persistent session memory across records. For hierarchical label sets (IUCN Threats, L2; Ecosystem Typology, L4), multiple sequential stateless calls were made per record, with the output of each stage passed as context to the next.
Item B.3: Relevant LLM configurations reported (as applicable), such as temperature, max tokens, seed, and number of runs: Explanation: List any configuration settings that may affect outputs, such as temperature, sampling parameters, penalties, stop sequences, number of completions or runs, quantization level, and reasoning-related settings.	The OpenAI Responses API was used in batch mode. Inference was stateless across records. For hierarchical classifications, later stages ran as additional stateless calls per record, with prior-stage outputs re-injected as context. Reasoning effort varied by task and was fixed before full-corpus application. Exact task-specific reasoning-effort settings, output-token limits, and any other non-default parameters are reported in the supplementary code repository.

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Reporting checklist for LLM use (continued).

Item	Answer
Item B.4: Customization: Explanation: Check and describe any modifications or extended capabilities incorporated into your LLM setup beyond standard inference, including fine-tuning, RAG, automated prompt optimization, web search integration, agentic workflows, or post-training refinements.	☒ Base model ☐ Fine-tuning ☐ RAG (retrieval-augmented) ☐ Automated prompt optimization ☐ Tool/function calling ☐ Web search ☐ Agentic workflows ☐ Other adaptations Description: Customization was limited to task-specific prompts, staged hierarchical inference for multi-level label sets, and a prebuilt GBIF Backbone Taxonomy cache used for taxonomic hierarchy lookup and name normalization in post-inference processing.
Item B.5: Did the LLM session(s) include persistent memory across interactions? Explanation: Indicate whether the LLM could “remember” previous conversations (i.e., had persistent memory). Unless such memory is disabled, there may also be spillover effects from other chat windows or prior conversations.	☐ Yes ☒ No ☐ N/A
Prompts	
Item C.1: Exact prompt(s) reported: Explanation: Whenever possible, include the exact text of prompts you used. If full prompts cannot be shared, include a redacted or representative example or link to the full prompt in a repository.	All prompts are available in the supplementary code repository.
Item C.2: System-wide instructions (if any): Explanation: Note any system-level instructions that guide the model’s general behavior (e.g., “You are a helpful assistant.”). These are commonly not directly visible but can be accessed through the API.	No single system-wide instruction was applied across all tasks. Each label set used a dedicated task-specific prompt.
Data inputs & privacy	

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Reporting checklist for LLM use (continued).

Item	Answer
Item D.1: Handling of personal or sensitive data (if any) (e.g., consent for data processing): Explanation: If any personal, sensitive, or identifiable data were processed, describe how they were handled in compliance with ethical standards and data protection laws. Clarify where data were stored or processed and how applicable legal/ethical requirements were met.	No personal or sensitive data were processed. All model inputs consisted of titles and abstracts of bibliographic records retrieved from the Web of Science Core Collection.
Validation & interpretation	
Item E.1: Human validation of LLM outputs: Explanation: If relevant, describe whether and how human reviewers examined the model's outputs, and the degree of independence they had in doing so. Specify the reviewers' roles and relevant expertise. Clarify what dimensions of performance were examined and what metrics were used.	☒ Yes ☐ No ☐ N/A Description: LLM labels were evaluated against the human-annotated records used in CC1 and CC2. Screening performance was assessed primarily using recall, and coding performance using precision, recall, F1, and Cohen's κ. Following full-corpus application, the authors conducted a post-hoc manual review of a sample of screened and coded records.

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Reporting checklist for LLM use (continued).

Item	Answer
Item E.2: Describe any relevant post-processing (e.g., filtering in case of format mismatches, unit conversions, etc.): Explanation: Describe any steps you took to clean or reformat LLM outputs. State how you handled inconsistent or unusable outputs and whether corrections were made with an automated script or manually.	Post-processing was fully automated and focused on parsing, schema harmonization, and label normalization. Outputs were requested in structured JSON format; responses were parsed into standardized records, with malformed JSON repaired using a JSON-repair utility where needed, and unrecoverable failures retained with explicit error fields. For evaluation, nested outputs were flattened into tabular fields, list-like predictions were deduplicated and normalized, and known label variants were canonicalized (including threat-label aliases, geography normalization to ISO 3166-1 alpha-3, and screening decision labels). Taxa predictions were enriched post hoc with higher-rank taxonomy fields using a prebuilt GBIF lookup cache.
Reproducibility	
Item F.1: Code/notebooks/scripts for LLM calls shared: Explanation: Indicate whether you have shared materials such as code, prompts, logs, or transcripts. Make sure sensitive information (e.g., API keys, private data) is removed. For code, make sure to add a README file.	☒ Yes ☐ No ☐ N/A Link/DOI: Code repository is provided
Competing interests	
Item G.1: Funding, support, or other relevant relationships (including in-kind access to compute or models, or professional affiliations): Explanation: Disclose any current or past funding, support, or other relevant relationships with entities that have a financial interest in LLMs. Disclose these relationships regardless of whether or not you believe they impacted the research.	☐ Yes. Description: ☒ No Link/DOI:
Optional items	

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Reporting checklist for LLM use (continued).

Item	Answer
Justification for the LLM choice along the following dimensions: Explanation: Briefly explain why you chose this model for your study and why it is suited to the research needs. Consider: performance, transparency, reproducibility, ethical considerations, as well as cost or ease-of-use if relevant.	■ Performance – Description: GPT-5-nano was selected for formal evaluation because it is suited to large-scale classification tasks. Its performance for screening and coding was assessed through CC1 and CC2. □ Transparency – Description: □ Reproducibility – Description: □ Ethical considerations – Description: ■ Others (e.g., cost, ease-of-use) – Description: GPT-5-nano provided a cost-effective option at the scale of this map (~2.4 million records), enabling large-scale automated processing without significant computational cost.
Discussion of the rationale for the prompt design: Explanation: Explain how you designed your prompts. Indicate whether you used a structured format, followed established prompt engineering guidelines, adapted prompts from earlier studies, or relied on automated prompt optimization tools. Clarify whether the design was iterative and whether few-shot examples were included.	Prompts followed a structured, task-specific format with explicit task definitions, decision procedures, controlled label vocabularies, and output constraints. Where applicable, label definitions and decision rules were adopted verbatim from established classification schemes used in this map (see protocol). Each prompt was paired with a structured output schema enforced at the API level. Prompts were refined iteratively on training and development splits; no automated prompt optimization was used.
Comparison against other methods/LLMs: Explanation: If comparisons were made between models, prompts, or methods, describe how the comparison was conducted, metrics used, and fairness of inputs. If alternatives were excluded, justify why.	■ Yes. Metrics: Specified in Item E.1 □ No □ N/A

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Reporting checklist for LLM use (continued).

Item	Answer
Training data leakage risks addressed: Explanation: Note whether risks of evaluation materials appearing in model training data were considered. Describe mitigation steps such as novelty checks or item restructuring.	☒ Yes: To prevent leakage within the study workflow, the held-out test split was not used for prompt development or refinement ☐ No ☐ N/A
Potential risk of bias or systematic differences in LLM behavior: Explanation: Reflect on whether the model may perform differently across groups, languages, or contexts in ways that could affect conclusions. Describe any bias checks or mitigation.	☒ Yes. Description: Potential systematic differences may occur across geographic contexts, taxonomic naming conventions, and article phrasing styles. No formal subgroup bias audit was conducted, but consistency was partially supported through controlled vocabularies, conservative decision rules, fallback labels (Unclear, Not Applicable), schema-constrained outputs, and post hoc normalization of geography and taxonomy outputs. ☐ No
Conversation transcripts: Explanation: For studies involving direct researcher/participant interaction with an LLM, provide anonymized transcripts or representative examples.	☐ Yes, shared without sensitive information. Location: ☐ No. Reason for not sharing: ☒ N/A. Description: No interaction involved.

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Reporting checklist for LLM use (continued).

Item	Answer
Relevant ethical implications of the research: Explanation: Discuss broader ethical considerations beyond procedural compliance, including participant well-being, fairness, safety, autonomy, and dual-use risks.	The main ethical implications of this work concern the use of LLMs for automated evidence synthesis. Misclassification or systematic differences in model behavior could bias downstream conclusions, particularly for brief or ambiguous abstracts. The map therefore used structured outputs, conservative rules, uncertainty labels, and evaluation against labeled data, and its outputs were interpreted in light of the reported validation results and methodological limitations.
Computational resources: Explanation: Report the computational resources required for the study. This may include API call counts, total tokens processed, financial costs, run-time, number and type of GPUs/CPUs used, and any other relevant compute metrics.	API Call Count: 4.15 Million Tokens Processed: 20.27 Billion (Input = 12.96 Billion, Output = 7.32 Billion) Financial Cost: € 1,400